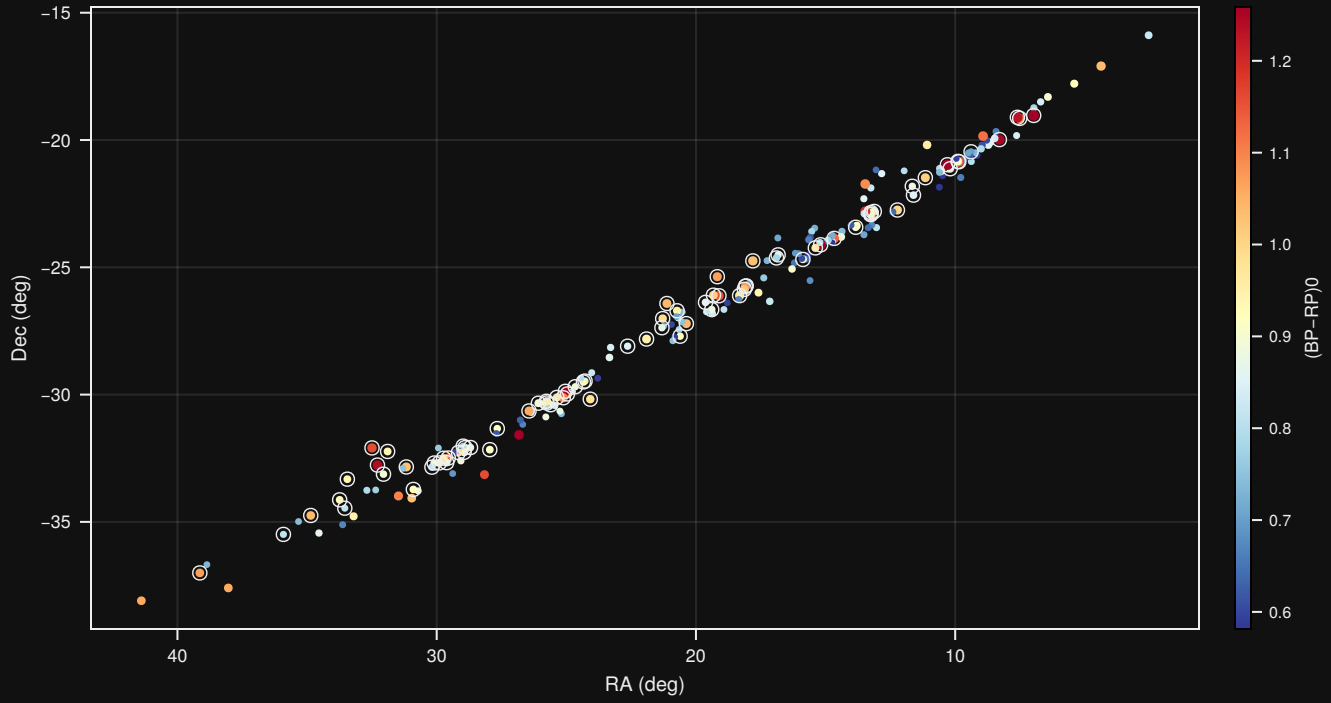
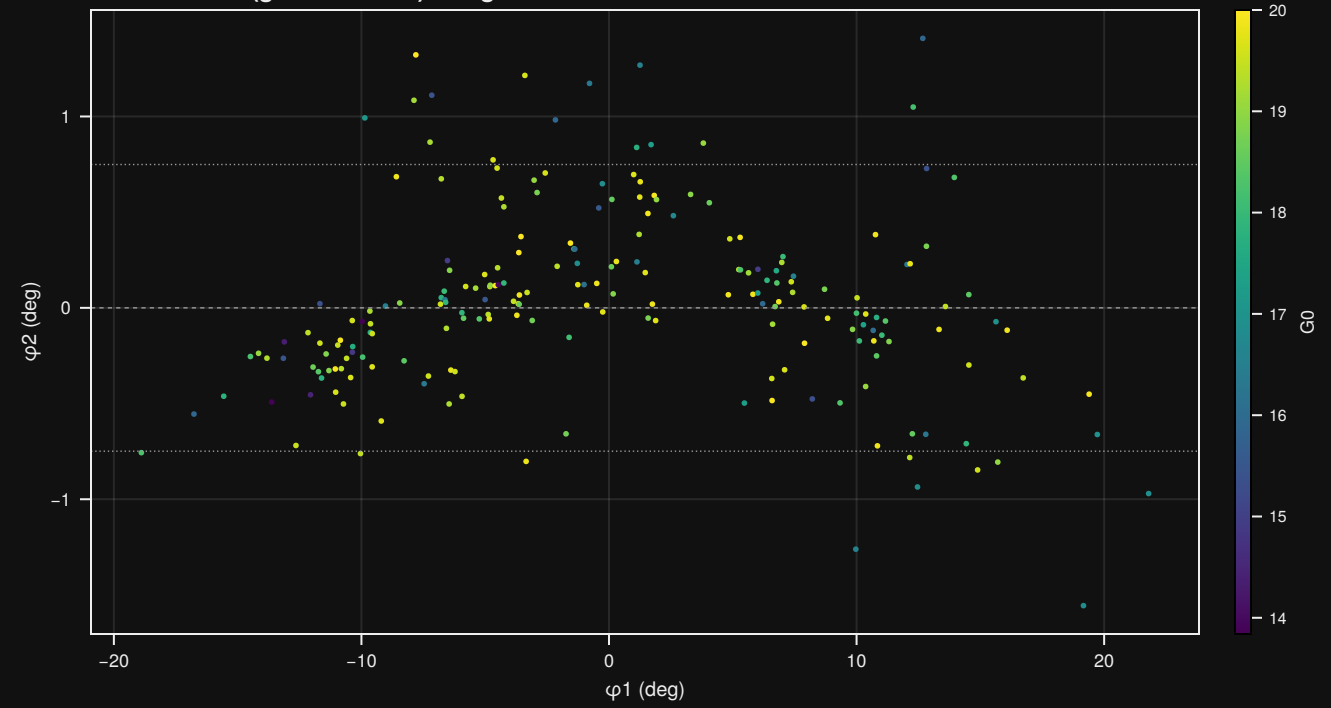
ATLAS

MEMBERSHIP	
Stars in catalog	208
with Gaia GSP-Phot	95 (46%)
with radial velocity	80
Stream id (sID)	4
GEOMETRY · great-circle fit	
Centroid RA, Dec	19.6, -26.0 deg
Centroid l, b	184.8, -82.7 deg
Pole RA, Dec	75.6, 47.6 deg
Length (φ1, 2–98%)	30.7 deg
Width σ(φ2), robust	0.37 deg
KINEMATICS	
pmRA* med ± σ	-0.16 ± 0.39 mas/yr
pmDE med ± σ	-0.96 ± 0.18 mas/yr
μφ1* med ± σ	0.39 ± 0.24 mas/yr
μφ2 med ± σ	-0.91 ± 0.31 mas/yr
Parallax (cat.) med	0.036 ± 0.185 mas
Parallax (wtd, ZP-corr.)	0.042 ± 0.012 mas
Distance	d = 23.8 ± 6.6 kpc
Physical length	12.78 kpc
RV median ± σ(MAD)	-99.3 ± 44.5 km/s
RV gradient dV/dφ1	3.94 km/s/deg
PHOTOMETRY · POPULATIONS	
G0 range (median)	13.8–20.0 (19.0)
(BP−RP)0 median	0.84
GSP-Phot [M/H] (n=95)	-2.16 ± 0.85
GSP-Phot Teff median	5296 K
RUWE > 1.4	0.0 %
PM clusters (largest)	12 (22 stars)
IBATA+2024 · TABLE 3	
n catalog | paper	208 | 208
n_RV catalog | paper	80 | 80
<[Fe/H]> paper	-2.20
Stellar mass (paper)	19000 Msun
Pole offset vs paper	0.0 deg

Sky map · catalog members (n = 208) · size ∝ brightness · ring = has RV

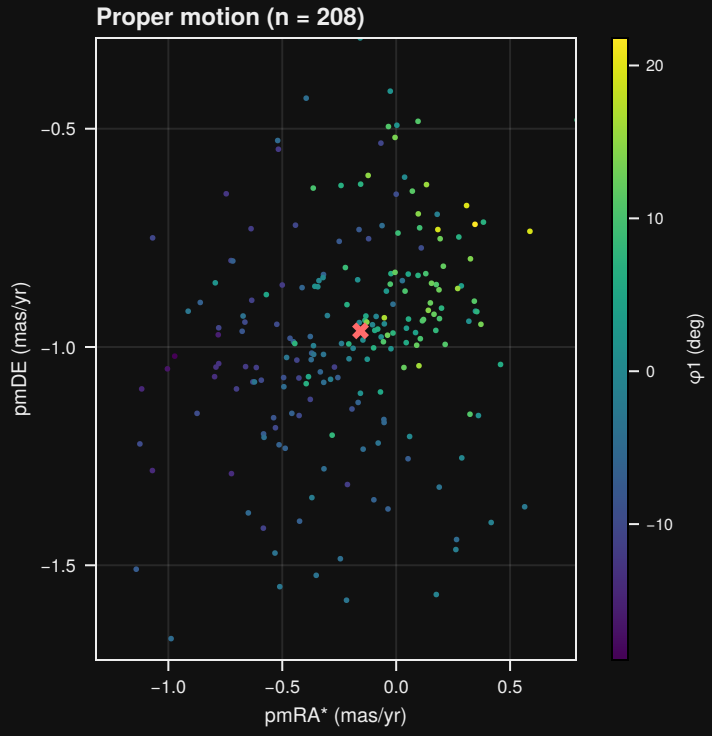


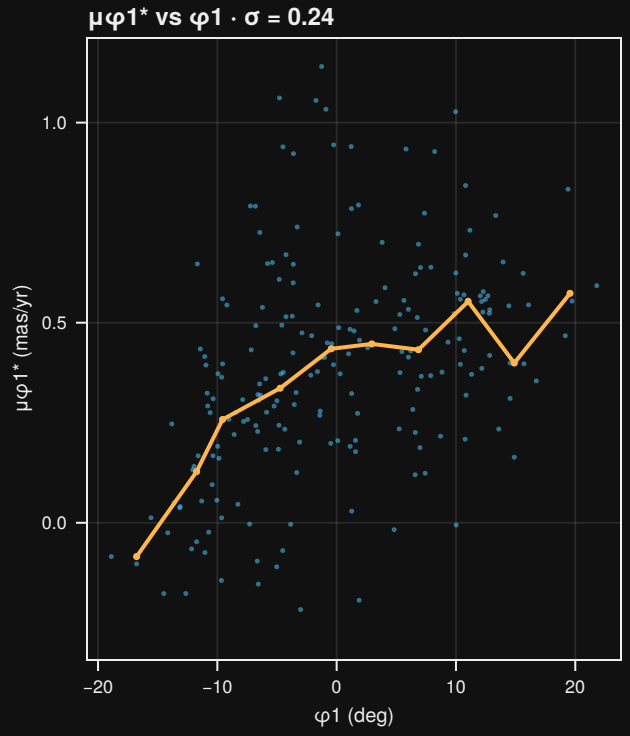
Stream frame (great-circle fit) · length 30.7° · width σ 0.37°

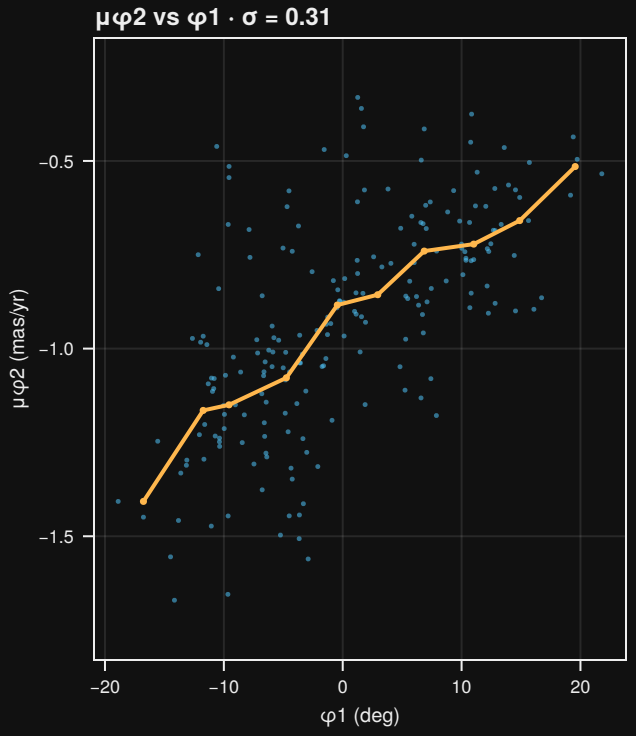


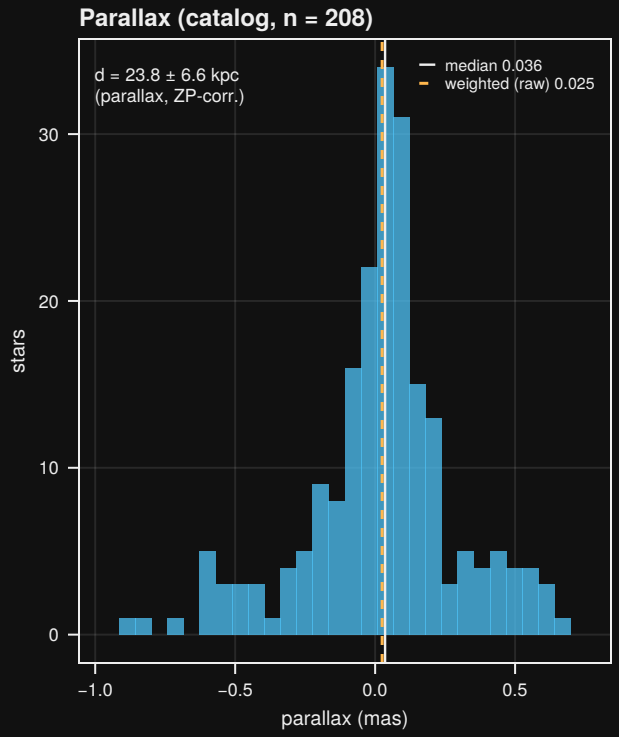
ATLAS

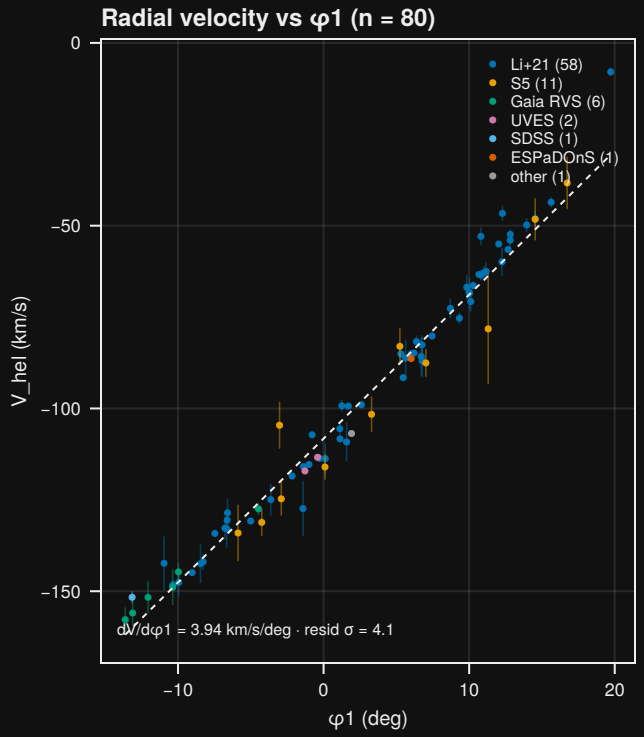
Kinematics · proper motion · parallax · radial velocity

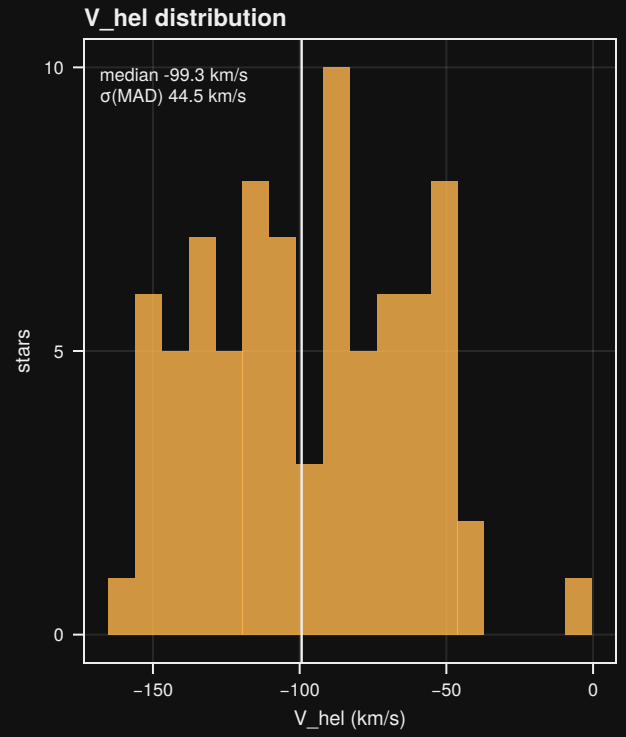






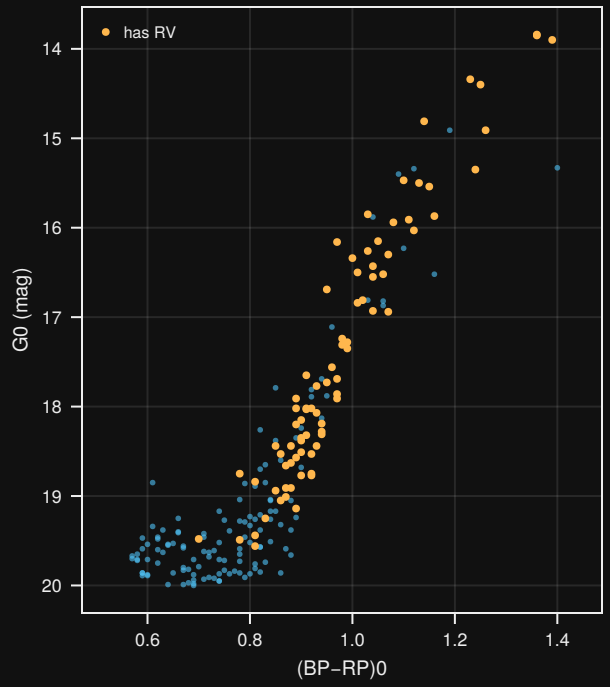




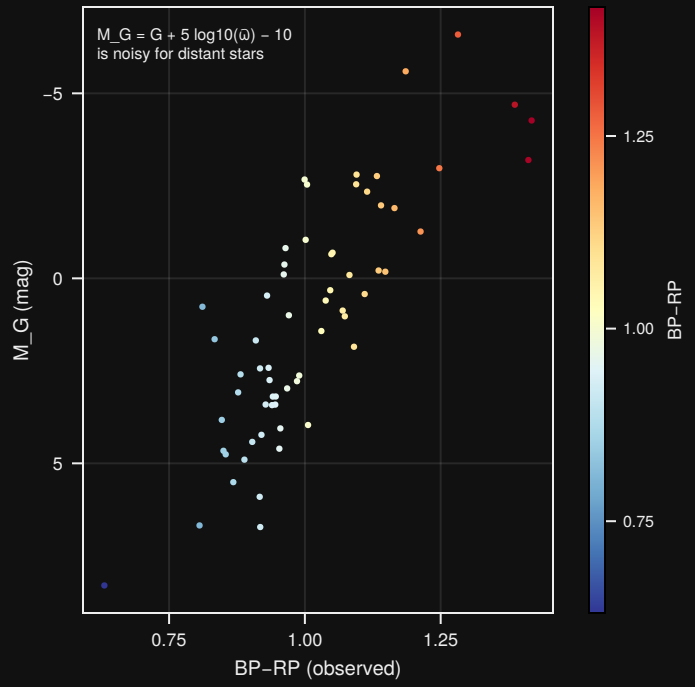


ATLAS

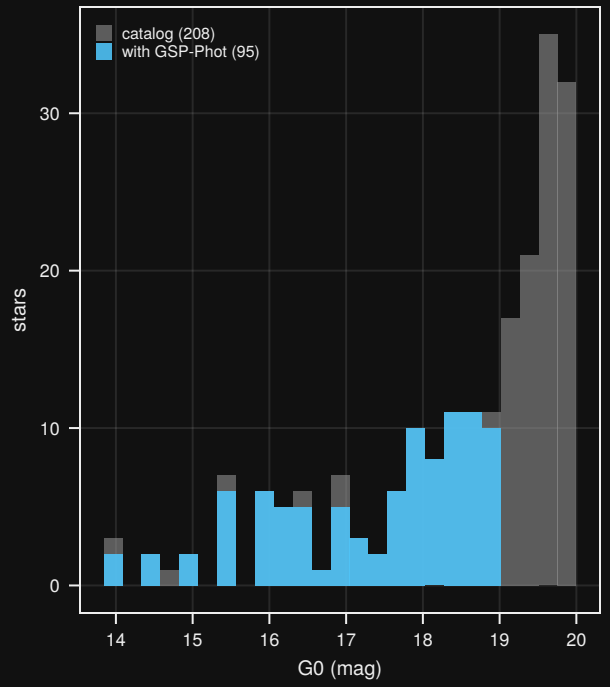
CMD · dereddened (n = 208)



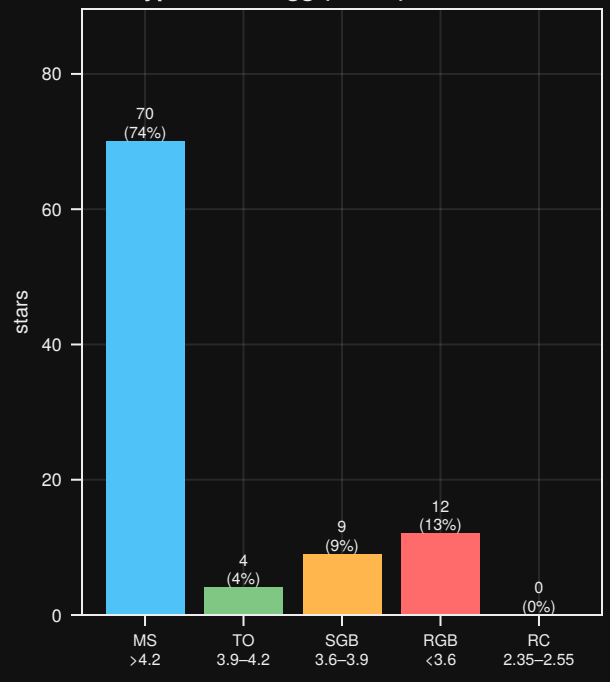
HR diagram · ϖ>0 (n = 63/95)



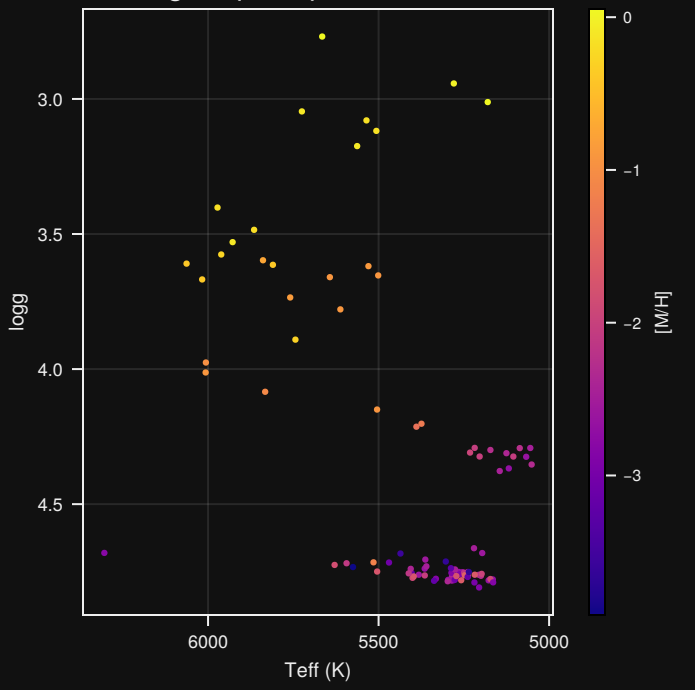
G0 & GSP-Phot coverage



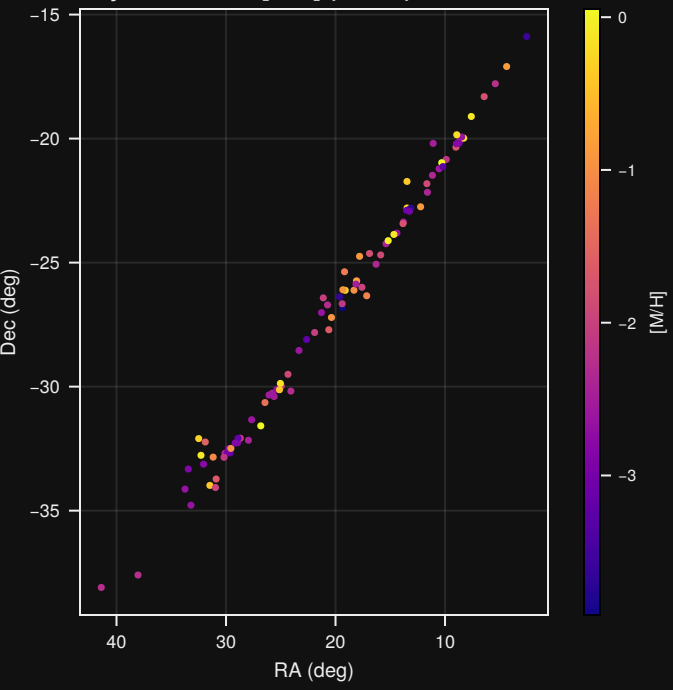
Star types from logg (n = 95)



Kiel diagram (n = 95)



Sky · GSP-Phot [M/H] (n = 95)

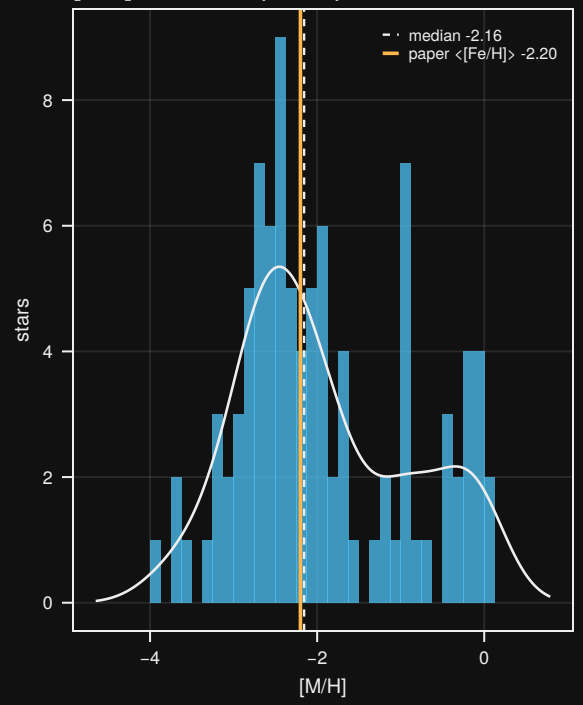


GSP-Phot (Gaia low-resolution BP/RP) parameters are prior-dominated for faint, distant halo stars (G ≥ 17): logg, [M/H] and ages are indicative only and typically biased toward dwarfs.

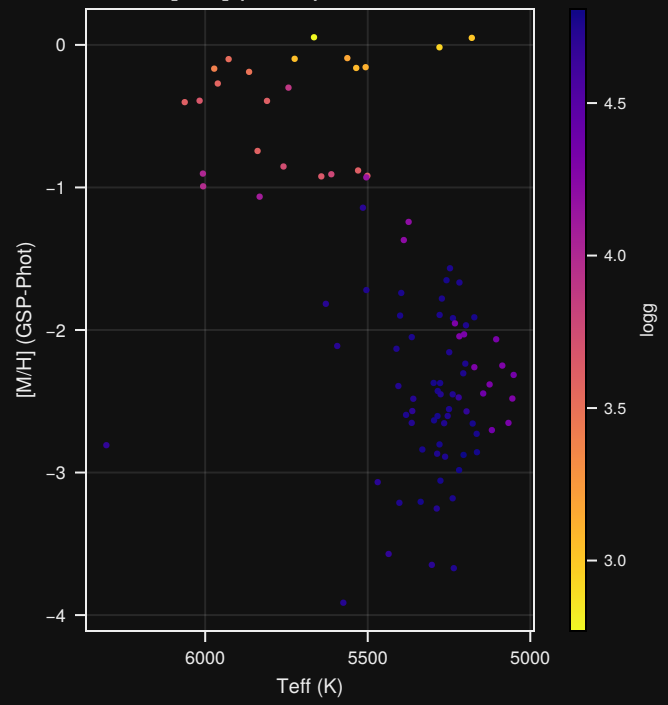
ATLAS

Metallicity · age · astrometric quality

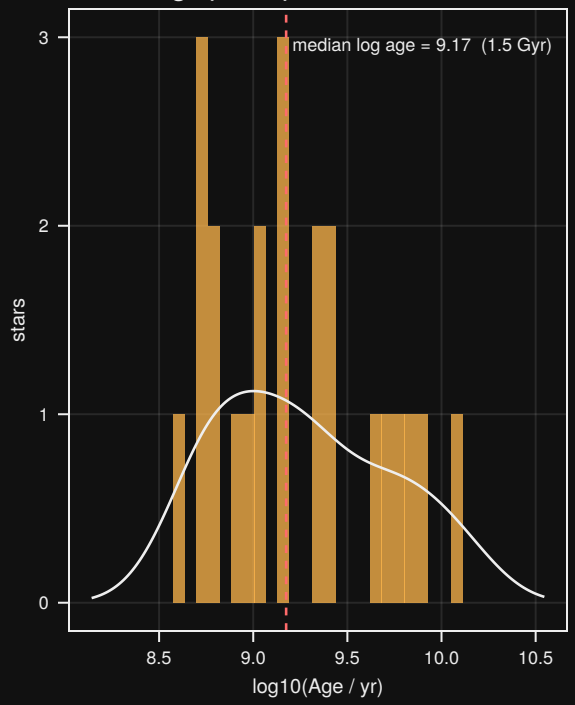
[M/H] GSP-Phot (n = 95)



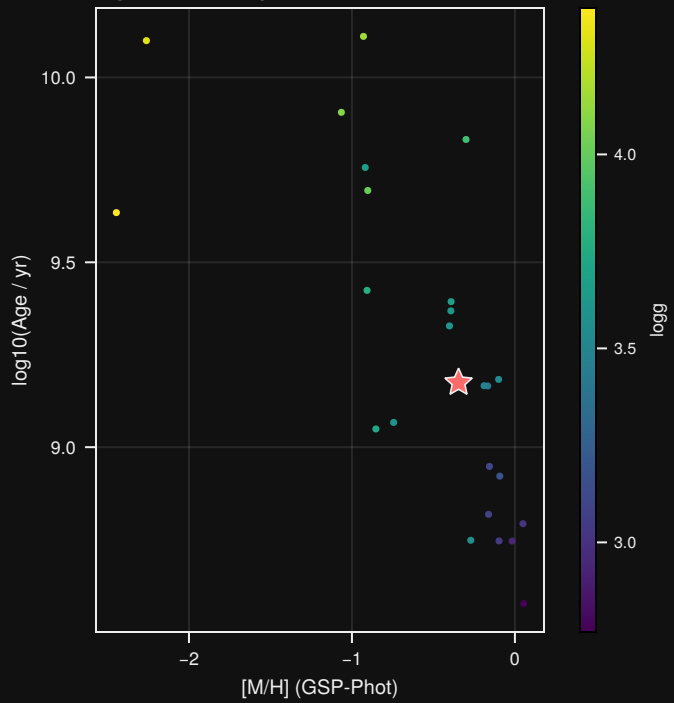
Teff vs [M/H] (n = 95)



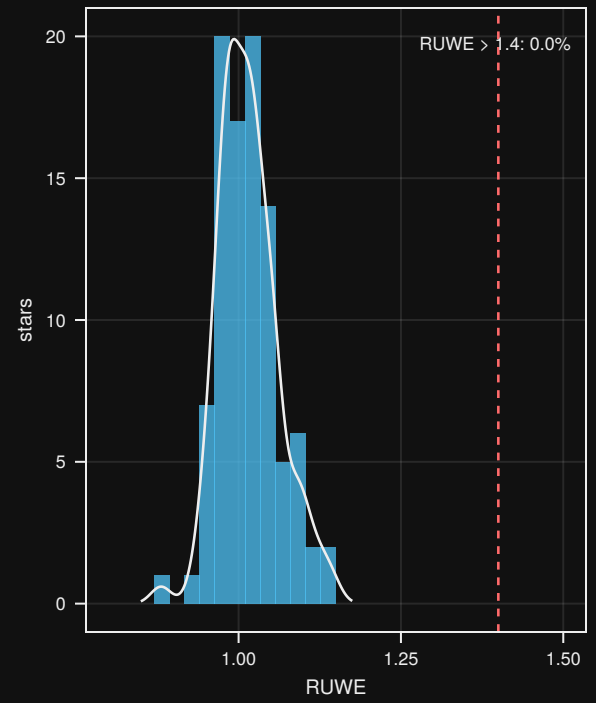
FLAME age (n = 24)



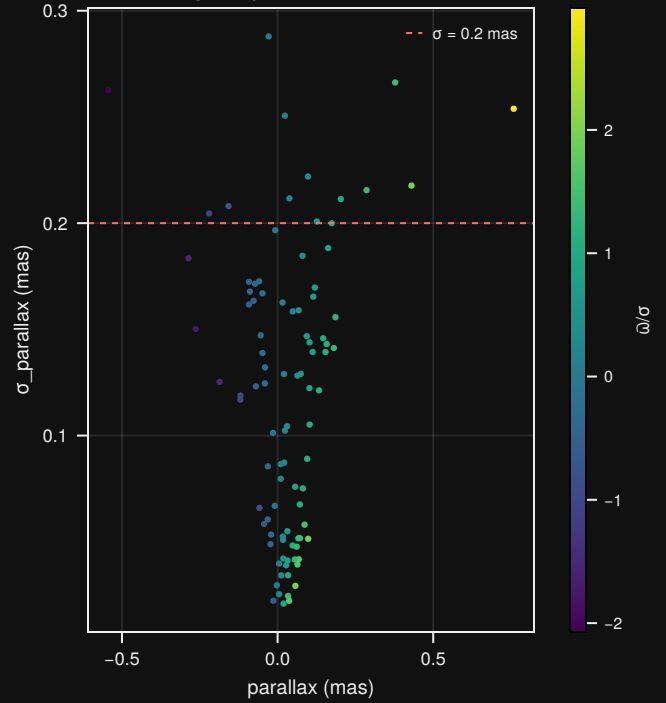
Age–metallicity (n = 24)



RUWE (n = 95)

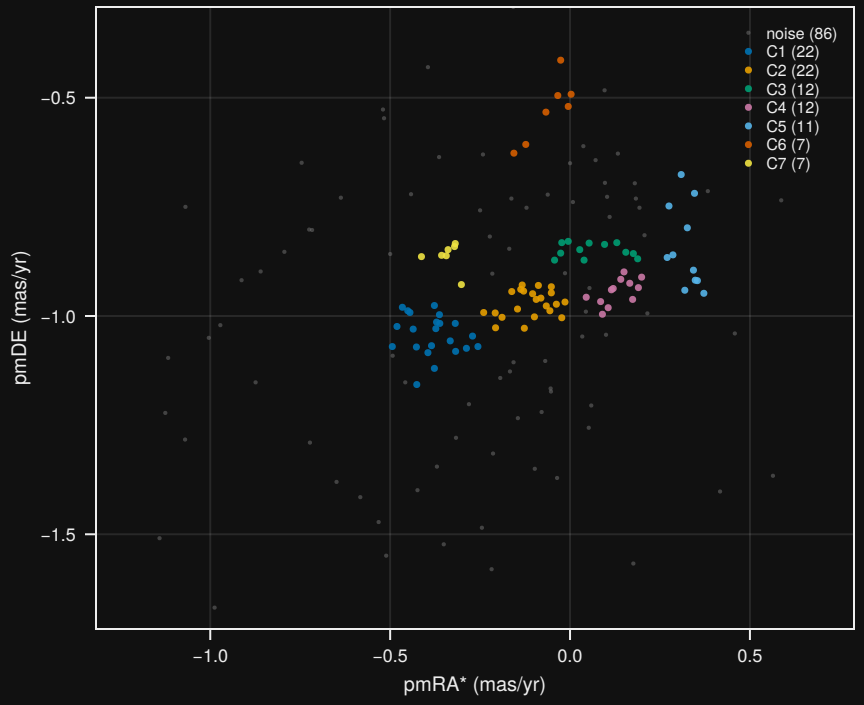


Parallax quality (n = 95)

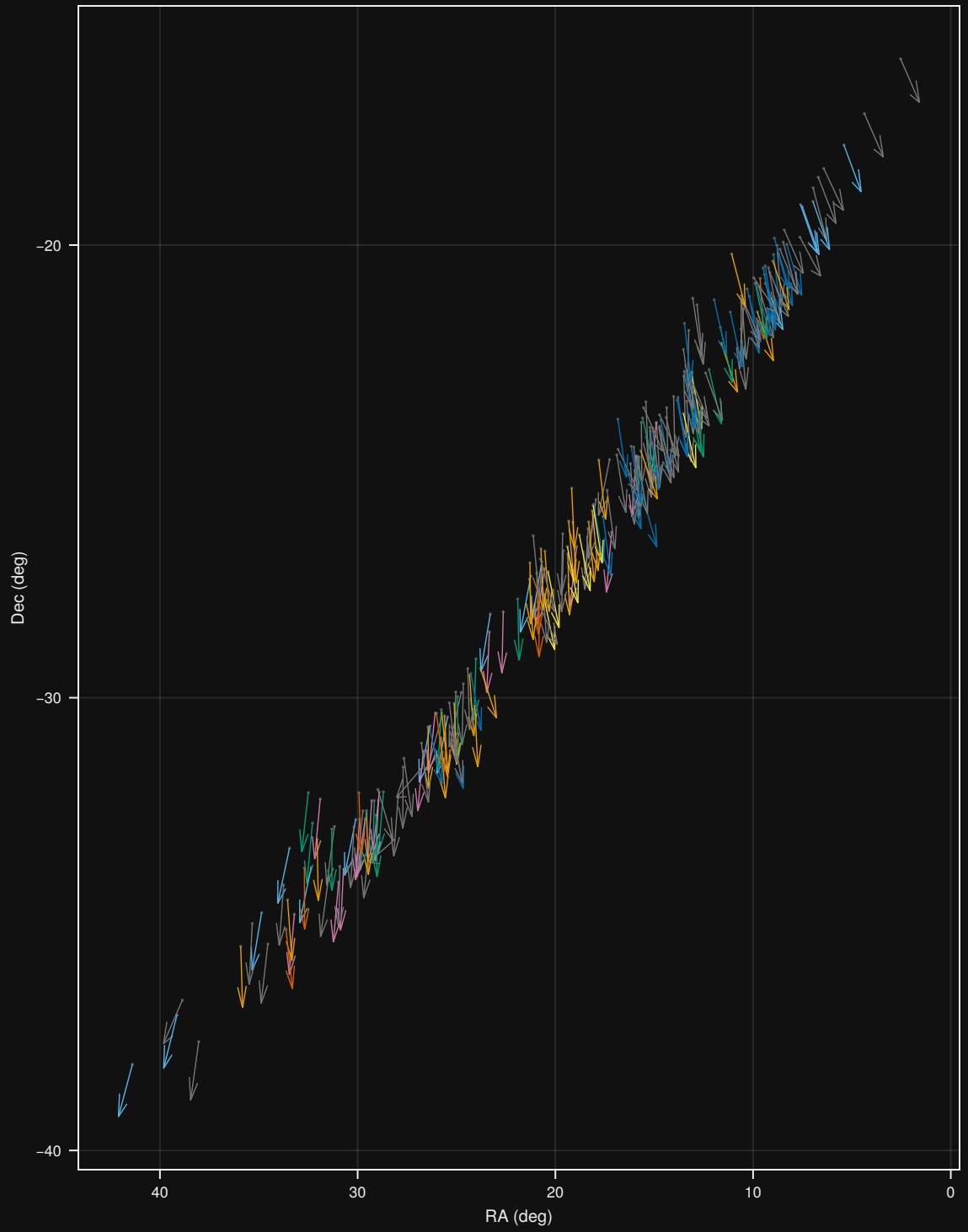


ATLAS

PM clustering · HDBSCAN · 12 cluster(s) · noise 41%



Proper-motion directions (arrows) coloured by cluster



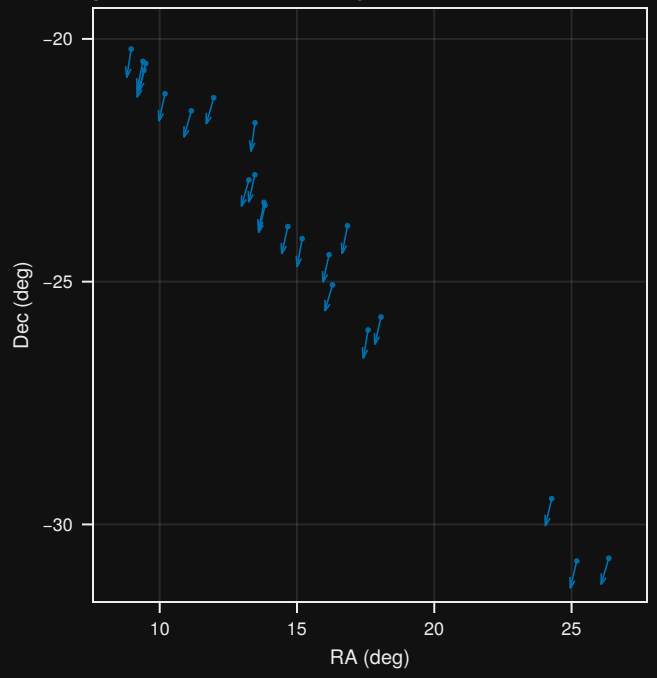
Cluster summary

id	n	pmRA	pmDE	σpmRA	[M/H]	n_RV
noise	86	-0.17	-0.99	0.40	-2.45	19
C1	22	-0.38	-1.04	0.08	-1.92	9
C2	22	-0.10	-0.97	0.07	-2.11	18
C3	12	0.05	-0.85	0.10	-2.04	8
C4	12	0.13	-0.94	0.06	-2.66	6
C5	11	0.33	-0.87	0.03	-2.37	4
C6	7	-0.03	-0.52	0.05	—	0
C7	7	-0.34	-0.86	0.03	-1.65	5
C8	6	-0.63	-1.08	0.03	-0.18	3

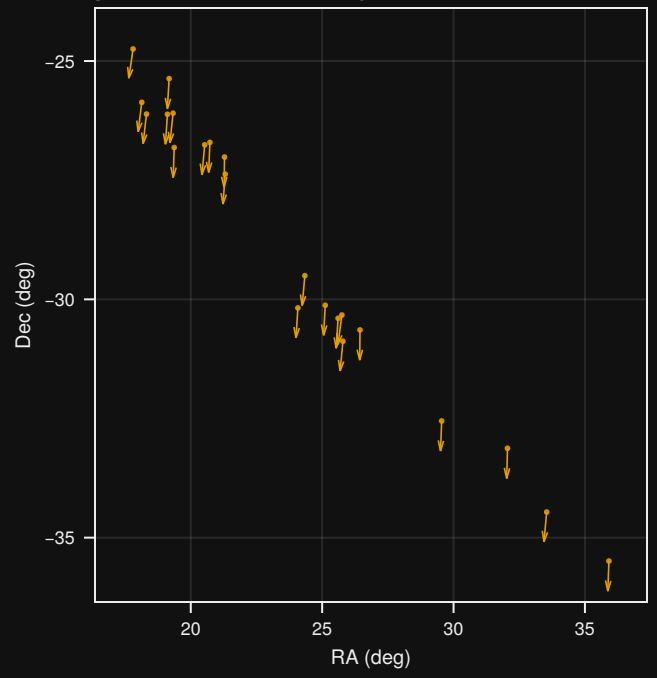
ATLAS

Per-cluster proper-motion vector maps

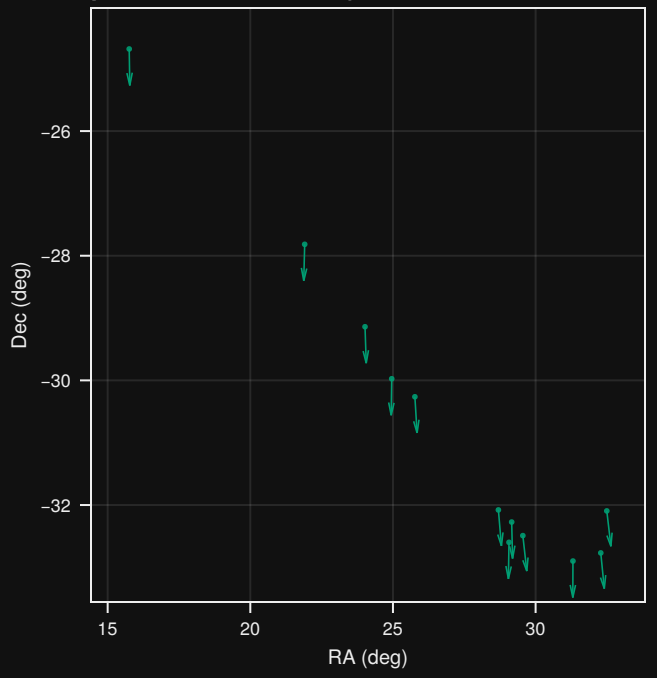
Cluster C1 · n = 22
pm = (-0.38, -1.04) mas/yr



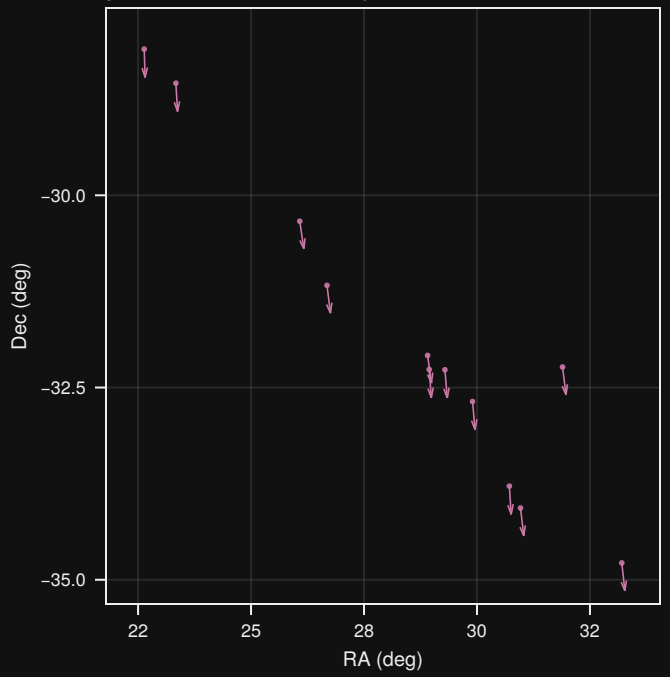
Cluster C2 · n = 22
pm = (-0.10, -0.97) mas/yr



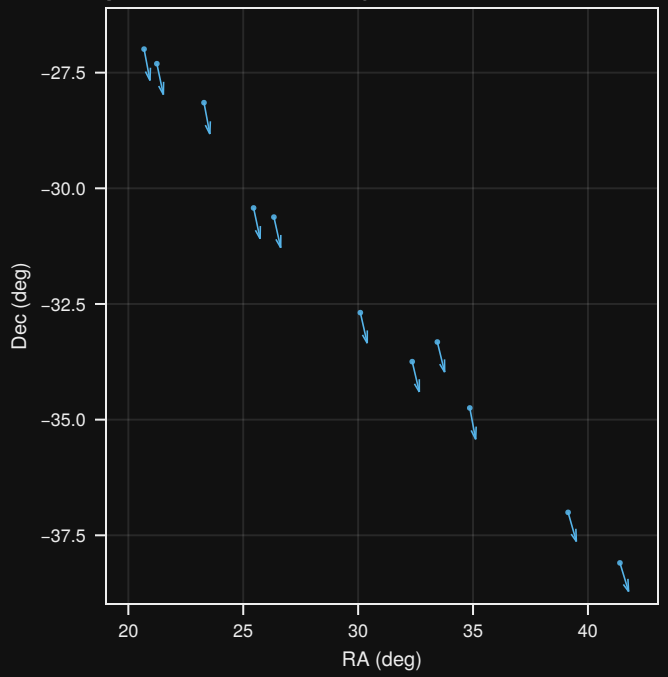
Cluster C3 · n = 12
pm = (0.05, -0.85) mas/yr



Cluster C4 · n = 12
pm = (0.13, -0.94) mas/yr



Cluster C5 · n = 11
pm = (0.33, -0.87) mas/yr



Noise · n = 86

